Dev Garg

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PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience designing, building, and deploying end-to-end machine learning solutions from concept to production. Expertise in developing and evaluating agentic AI systems with proven ability to translate business needs into scalable AI solutions.

EXPERIENCE

Encando.AI

Founding/Senior Software Engineer — *College Station, TX* · **May 2025 - Present**

- **Architected a scalable, AI-first LMS backend** from zero-to-one, scaling to **5,000+ active students** in 5 months while optimizing architecture to reduce median latency by **95% (1.4s to 67ms)** and improve P95 latency by **3.5x**.
- Developed and optimized a **multi-stage agentic RAG pipeline** using LangChain, Pinecone vector databases and OpenAI API to automate-complex workflows and boosting response relevance by **20%** (measured via **Ragas framework**).
- **Led a team of 5 engineers** to establish a high-velocity CI/CD culture, enabling **reliable daily releases** by implementing rigorous code review standards, agile workflows and automated testing pipelines.
- Established **end-to-end ownership** of the platform, analyzed and optimized system efficiency, reducing monthly AWS infrastructure costs by 15% and cutting AI inference costs by **35%** through **token-efficient prompt design**.
- **Designed and migrated** the monolithic backend to a highly available, Multi-AZ architecture using Application Load Balancers, eliminating a single point of failure and increasing system throughput by **50% (18 to 27 RPS)** under load.

Societe Generale GSC

Software Engineer — *Bengaluru, India* · **May 2021 - Dec 2023**

- Led the development of **scalable data processing** systems for credit risk analysis, enabling high-volume data ingestion for downstream analytics by utilizing **Big Data technologies** like Apache Spark and Kafka.
- Accelerated system validation cycle times by 50% and reduced quarterly compute costs by **€20,000** by building an automated regression testing suite and designing auto-scaling data pipelines.
- Built and optimized **distributed data workflows** using Java, Spring Boot, SQL, and Scala, supporting critical analytics capabilities for real-time risk management applications.

Software Engineer Intern — *Remote* · **May 2020 - Jun 2020**

- Designed and delivered an MVP for an incident resolution system, improving operational response times by 38% by **leveraging NLP to recommend solutions** from historical data, collaborating with cross-functional teams.

EDUCATION

Texas A&M University — **Jan 2024 - May 2025**

Master of Computer Science, Computer Science

- **GPA:** 4.0/4.0
- **Coursework:** Large Language Models, Software Engineering, Deep Learning, Info Storage and Retrieval

Indian Institute of Technology, BHU — **2017 - 2021**

Bachelor of Technology, Electronics Engineering

- **GPA:** 8.78/10.0

TECHNICAL SKILLS

- **Languages**: Python, Java, JavaScript, C++, SQL, TypeScript
- **Frameworks**: PyTorch, LangGraph, OpenAI API, Scikit-learn, LangChain, Spring Boot, Apache Spark, React
- **Platforms**: AWS, Kubernetes, Docker, Azure, GCP, ElasticSearch
- **System Design & MLOps**: System Architecture, Distributed Systems, Agile Methodologies, CI/CD
- **Certifications**: Google Cloud Professional MLE, Machine learning Spec., MLOps Spec., GenAI with LLMs

PROJECTS

PaperFormer: A Citation-Graph Enhanced Language Model for Scientific Applications. | [Under Review]

- Co-developed a citation-aware **LLM based on LLaMA 3.2**, integrating full-text citation contexts via LoRA fine-tuning and custom weights, achieving a 51% perplexity reduction in causal language modeling and **SOTA** ROUGE-1 (47.85) in paper summarization on the SSN dataset.
- Engineered a distributed data pipeline processing 21K papers and 1.13M citations using Ray, enabling **scalable AI-driven** review generation and reducing processing time by 60%

News Aggregation and Recommendation System

- Designed and implemented a **scalable event-driven microservices** architecture for a real-time news aggregation and recommendation platform, using FastAPI, Kubernetes, and Kafka for high availability and fault tolerance.
- Designed and implemented an automated **MLOps pipeline** with a model monitoring system to track drift and bias, enabling automatic retraining triggers to maintain system quality.
- Developed a service discovery mechanism leveraging Kubernetes DNS-based discovery and API Gateway load balancing, enabling dynamic scaling and resilience in a multi-service environment. Optimized distributed data processing pipelines using PySpark, improving ingestion and clustering performance for high-volume news data.